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# 15 Essential Skills for Cybersecurity Analysts in 2026

Written by Coursera Staff • Updated on Dec 8, 2025
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Getting a job as a cybersecurity analyst often means having the right set of technical and workplace skills to navigate the demands of this important role. Here are 15 cybersecurity skills you can develop or strengthen.



## Key takeaways

- Working as a cybersecurity analyst requires a range of technical skills, such as intrusion detection, to help you keep an organization safe from cyber threats.
- Being a cybersecurity analyst also requires workplace skills, such as communication, to document incident reports, among other tasks.
- You can enhance your cybersecurity skills through courses and workshops, and by staying informed about current trends.

Learn more about the key skills you'll need to succeed as a cybersecurity analyst. Afterward, build many of these skills in the Microsoft Cybersecurity Analyst Professional Certificate, where you'll learn how to understand the vulnerabilities of a company's network and mitigate attacks on network infrastructures.




professional certificate

### Microsoft Cybersecurity Analyst

Launch your career as a cybersecurity analyst. Build job-ready skills – and must-have AI skills – for an in-demand career. Earn a credential from Microsoft. No prior experience required.

4.7

(3,073 ratings)

147,620 already enrolled

Beginner level

Learn More

Average time: 6 month(s)

Learn at your own pace

Skills you'll build:

Security Strategy, Penetration Testing, Cloud Security, Intrusion Detection and Prevention, Active Directory, Authentications, Operating Systems, Compliance Management, Data Storage Technologies, Cloud Computing, Security Information and Event Management (SIEM), Cryptography, Threat Modeling, Azure Active Directory, Computer Security Incident Management, Cyber Security Strategy, Encryption, Vulnerability Management, MITRE ATT&CK Framework, Network Security, Virtual Machines, Firewall, Microsoft Azure, Computer Networking, General Networking, Virtualization, Digital Transformation, Artificial Intelligence and Machine Learning (AI/ML), Network Infrastructure, Virtualization and Virtual Machines, Cloud Services, Cybersecurity, Networking Hardware, Network Protocols, Network Monitoring, Control Objectives for Information and Related Technology (COBIT), Data Security, General Data Protection Regulation (GDPR), Business Continuity, Information Privacy, Data Management, Threat Detection, Cyber Governance, Business Continuity Planning, Regulation and Legal Compliance, Security Management, Data Mapping, Records Management, Law, Regulation, and Compliance, Threat Management, Regulatory Compliance, Multi-Factor Authentication, Generative AI, Cyber Risk, Identity and Access Management, Security Software, Data Governance, Distributed Denial-Of-Service (DDoS) Attacks, Incident Response, Event Management, Cyber Security Policies, Malware Protection, Cyber Attacks, Endpoint Security, Security Controls, Authorization (Computing), Patch Management, Application Security, Role-Based Access Control (RBAC), Single Sign-On (SSO), Enterprise Security, Vulnerability Scanning, Command-Line Interface, Virtual Private Networks (VPN), Windows

PowerShell, System Testing, Test Tools, Security Testing, Internet Of Things, IT Security Architecture, Asset Management, IT Infrastructure, Computer Architecture, Productivity Software, Email Security

# 15 essential skills to succeed as a cybersecurity analyst

Cybersecurity analysts use a combination of technical and workplace skills to assess vulnerabilities and respond to security incidents.

## 10 technical skills for cybersecurity analysts

A cybersecurity analyst's technical skills are the foundation of their career. Below, we cover ten of the most essential technical skills that you'll need to thrive in the field.

### 1. Scripting

Scripting is a type of coding that enables you to create a program that performs a specific task. The difference is that coding is static, whereas scripts can animate images and text. Knowing how to build tools and automate repetitive tasks with languages like Python or PowerShell empowers you to become a more efficient analyst. Python, in particular, ranks among the most prevalent languages in cybersecurity. As a bonus, it’s also among the easiest to learn.

RICE

specialization

### Introduction to Scripting in Python

Launch Your Career in Python Programming. Master the core concepts of scripting in Python to enable you to solve practical problems.

4.7

(4,370 ratings)

52,716 already enrolled

Beginner level

Learn More

Average time: 1 month(s)

Learn at your own pace

Skills you'll build:

Data Manipulation, Data Analysis, Development Environment, File I/O, Data Processing, Data Mapping, Programming Principles, Data Structures, Data Visualization, Computational Logic, Plot (Graphics), Code Reusability, Data Access, Software Installation, Python Programming, Package and Software Management, Data Cleansing, Scripting, Data Storage, Data Visualization Software, Computer Programming, Program Development, Debugging, Data Import/Export, Data Management, Technical Documentation, Data Integration, Software Documentation

**2. Controls and frameworks**

A cybersecurity framework provides a collection of best practices, policies, tools, and security protocols designed to help secure an organization's data and business operations. A control is a measure your company uses to protect itself from vulnerabilities and attacks.

The framework you use will vary depending on your organization and industry. You may find it helpful to familiarize yourself with some of the most common cybersecurity frameworks, including:

- National Institute of Standards and Technology (NIST)
- International Organization for Standardization (ISO)
- Center for Information Security (CIS)
- System and Organization Controls 2 (SOC 2)

**3. Intrusion detection**

As a cybersecurity analyst, a significant part of your job involves monitoring network activity for potential intrusions. Knowing how to utilize intrusion detection software—such as security information and event management (SIEM) products, intrusion detection systems (IDS), and intrusion prevention systems (IPS)—enables you to quickly identify suspicious activity or security violations.




specialization

## Intrusion Detection

Master Intrusion Detection for Cybersecurity . Gain expertise in intrusion detection systems and advanced network analysis to effectively respond to cybersecurity threats.

4.8

(10 ratings)

1,040 already enrolled

Intermediate level

Learn More

Average time: 4 month(s)

Learn at your own pace

Skills you'll build:

Machine Learning Methods, Threat Detection, Cyber Security Policies, Distributed Denial-Of-Service (DDoS) Attacks, Artificial Neural Networks, Cloud Security, Incident Response, Network Analysis, Cybersecurity, Applied Machine Learning, Machine Learning Algorithms, Intrusion Detection and Prevention, Anomaly Detection, Network Architecture, Data Analysis, Virtual Machines, Machine Learning, Model Evaluation, Computer Security Incident Management

**4. Network security control**

Many cybersecurity attacks occur across a network of interconnected devices. The same technologies that enable companies to collaborate can also create security vulnerabilities. To keep an organization secure, you'll need an understanding of wired and wireless networks and how to secure them.




course

## Introduction to Networking and Cloud Computing

This course forms part of a series of courses that offers a good starting point for a career in cybersecurity. It will help you gain knowledge and skills ...

4.8

(570 ratings)

54,446 already enrolled

Beginner level

Learn More

Average time: 24 hour(s)

Learn at your own pace

Skills you'll build:

Digital Transformation, Artificial Intelligence and Machine Learning (AI/ML), Cloud Computing, Network Infrastructure, Virtual Machines, Virtualization and Virtual Machines, Network Security, Cloud Services, Firewall, Microsoft Azure, Computer Networking, Cybersecurity, Networking Hardware, General Networking, Virtualization, Network Protocols, Network Monitoring




**5. Operating systems**

Security threats exist across all operating systems, both on computers and mobile devices. Set yourself up for success as a security analyst by building a deep familiarity with macOS, Windows, Linux, and their command-line interfaces. You may also find it helpful to study the threats and vulnerabilities associated with mobile operating systems, such as iOS and Android.

**6. Incident response**

While prevention is the primary goal of cybersecurity, responding promptly to security incidents is crucial to minimize damage and loss. Effective incident handling requires familiarity with your organization's incident response plan, as well as skills in digital forensics and malware analysis.

**7. Cloud**

As more businesses migrate to cloud environments, professionals with cloud expertise are in high demand. According to data from GIAC Certifications, possessing cloud security skills can result in a salary premium of more than $15,000 [1].

Microsoft

professional certificate

## Microsoft Cloud Support Associate

Launch your career as a Cloud Support Associate. Build job-ready skills for an in-demand career in information technology in as little as 6 months. No prior experience required to get started.

4.7

(1,685 ratings)

19,010 already enrolled

Beginner level

Learn More

Average time: 5 month(s)

Learn at your own pace

Skills you'll build:

Hybrid Cloud Computing, Microsoft Azure, Collaborative Software, Cloud Services, Cloud Solutions, Load Balancing, Microsoft Power Automate/Flow, Authorization (Computing), General Networking, Cloud Deployment, Authentications, Microsoft Teams, Hardware Troubleshooting, Security Controls, Generative AI, IT Automation, Dashboard Creation, Cloud Computing, Computer Networking, Network Protocols, Network Troubleshooting, Microsoft Windows, Technical Support, System Monitoring, Technical Documentation, Technical Communication, Operating Systems, Computer Hardware, Mac OS, Cloud Security, Disaster Recovery, Patch Management, Microsoft Copilot, Cloud Management, Virtual Machines, Microsoft Excel, Data Loss Prevention, Microsoft PowerPoint, Microsoft Outlook, Microsoft 365, Endpoint Security, Microsoft Office, Cloud-Based Integration, Productivity Software, Data Security, Cloud Storage, Virtual Networking, System Configuration, Network Security, Cloud Infrastructure, Role-Based Access Control (RBAC), Identity and Access Management, Application Deployment, Kubernetes, Network Architecture, Cloud Computing Architecture, Network Administration, Cloud Hosting, Network Monitoring, Data Storage, Data Management, Software Installation, Hardening, Information Systems Security, Servers, Business Software, Cybersecurity, Computer Architecture, AI literacy, Computer Systems, Encryption, Multi-Factor Authentication, Virtual Private Networks (VPN), Network Performance Management, Information Privacy, Solution Design, Query Languages, Incident Response, Resource Management, Event Monitoring, Performance Tuning, Azure Active Directory, Infrastructure as Code (IaC), Windows PowerShell, Network Planning And Design, Command-Line Interface, OSI Models, Firewall, TCP/IP, Virtualization, Network Model, Networking Hardware, Network Infrastructure, Technical Support and Services

**8. DevOps**

Security risks often exist within applications themselves. Increasingly, companies are incorporating a security focus into their software development and operations (DevOps) phases to ensure that applications are secure from the outset.

**9. Threat knowledge**

“Know thy enemy,” General Sun Tzu once said. You can be a more effective cybersecurity analyst by keeping up-to-date on the threat landscape. If you’re new to the field, get started with the Open Web Application Security Project (OWASP) Top 10—a document that outlines the top 10 web application security risks.



course

### Threat Analysis

If you are an associate-level cybersecurity analyst who is working in security operation centers, this course will help you understand Incident Analysis in ...

4.9

(36 ratings)

8,189 already enrolled

Learn More

Average time: 26 hour(s)

Learn at your own pace

Skills you'll build:

Network Protocols, Network Analysis, Exploitation techniques, Threat Detection, MITRE ATT&CK Framework, Network Security, Cyber Threat Intelligence, Anomaly Detection, Computer Security Incident Management

**10. Regulatory guidelines**

Cybersecurity must protect an organization from attacks, theft, and loss, while also complying with industry regulations. If you're working for a company that does business around the globe, familiarity with General Data Protection Regulation (GDPR) could be beneficial. Cybersecurity analysts in the health care industry will need to understand how to comply with the Health Insurance Portability and Accountability Act (HIPPA)—a US federal law that helps protect the privacy of medical records. Some states within the US have their own privacy laws as well.

**Read more:** Cybersecurity Terms: A to Z Glossary

## 5 workplace skills to succeed as a cybersecurity analyst

Workplace skills are abilities you rely on to actually get your work done, collaborate with others, and accomplish your professional goals. Below, we explore some of the most important workplace skills you'll need to become the best cybersecurity analyst possible.

**1. Communication**

Both written and verbal communication play crucial roles in cybersecurity. As an analyst, you may need to communicate technical concepts to individuals without a technical background, such as executives or legal teams. You may also be asked to write incident reports, where you'll have to document what you did in a concise and clear manner.

**2. Collaboration**

As a cybersecurity analyst, you'll likely work with a larger security team of other cybersecurity professionals. You may also need to collaborate with other teams within your company (legal, IT, public relations) or share your findings with other organizations or the greater cybersecurity community.

**3. Risk management**

Your ability to think through what could possibly go wrong, assess the severity of threats, and gauge the potential impact empowers you to focus your energy on the tasks where you'll have the biggest impact.




specialization

## Introduction to Cybersecurity & Risk Management

Gain Skills in Cybersecurity and Risk Management. The three interconnected courses in this specialization cover security governance and compliance strategies, foundational risk management techniques, as well as personnel and third-party security measures.

4.6

(354 ratings)

13,266 already enrolled

Beginner level

Learn More

Average time: 1 month(s)

Learn at your own pace

Skills you'll build:

Cyber Security Policies, Business Continuity, Business Continuity Planning, Regulatory Compliance, Cyber Governance, Compliance Management, Threat Modeling, Security Awareness, Cyber Risk, Security Strategy, Cybersecurity, Risk Analysis, Risk Management, Risk Mitigation, Supplier Risk Management, Security Management, Computer Security Awareness Training, Asset Protection, Risk Management Framework, Risk

Control, Regulation and Legal Compliance, Governance, Procedure Development, Regulatory Requirements, Law, Regulation, and Compliance, Cyber Security Strategy, Security Controls, Due Diligence, Human Factors (Security)

**4. Adaptability**

Cyber criminals are constantly adjusting and enhancing their attacks. Technology continues to advance, introducing new vulnerabilities. Adopting the mindset of a lifelong learner can help you keep up with (or stay one step ahead of) these changes.

**5. Critical thinking**

Working in cybersecurity sometimes means making high-stakes decisions about your organization's security. Developing your critical thinking skills can help you to:

- Ask the right questions
- Evaluate and assess data
- Identify your assumptions
- Consider alternatives
- Understand context
- Draw data-driven conclusions

## How to improve cybersecurity skills

Here are some options for building cybersecurity skills:

### 1. Take classes.

Whether you're learning cybersecurity fundamentals or more advanced skills, enrolling in a course adds structure to your learning. Many courses that target specific technical skills, like network security or incident response, may also give you opportunities to practice workplace skills, like technical writing and collaboration.

**Read more:** Is Cybersecurity Hard to Learn? 9 Tips for Success

### 2. Earn a certification or certificate.

Beyond a one-off course, you can find cybersecurity certifications or certificate programs to both gain new knowledge and formalize what you already know. These programs encompass a range of topics, including network security, ethical hacking, risk management, and compliance.

While certifications typically involve rigorous exams and may require work experience, certificate programs often offer more flexibility and can serve as stepping stones for newcomers or those seeking to specialize in

specific areas of cybersecurity. Either can be beneficial to add to your resume, showing potential employers the extra effort you've put into developing your skill set.




course

### CompTIA CYSA+ CS0-003: Security Operations

This course delves into the critical aspects of security operations, equipping you with the skills to proactively protect your organization's digital ...

4.3

(14 ratings)

4,530 already enrolled

Intermediate level

Learn More

Average time: 7 hour(s)

Learn at your own pace

Skills you'll build:

Cyber Threat Intelligence, Encryption, Security Strategy, Vulnerability Management, Cyber Threat Hunting, Cyber Attacks, Vulnerability Assessments, Security Controls, Cloud Security, Vulnerability, Cyber Security Strategy, Incident Response, Systems Architecture, Cyber Operations, MITRE ATT&CK Framework, Threat Detection, Cybersecurity, Network Security, Security Information and Event Management (SIEM), Configuration Management

## 3. Stay up to date on the latest trends.

Keeping your technical cybersecurity skills up to date with the current threat and technology landscape can help you build confidence as a security analyst and give you a competitive advantage in your job search. Pay attention to cybersecurity trends throughout the year, and look for resources to engage with other professionals. Here are some ideas to get you started:

- SANS StormCast**:** Subscribe to this daily podcast for five to 10-minute episodes covering the latest security threats.
- Security Now**:** Steve Gibson who coined the term “spyware” discusses current topics in cybersecurity on this weekly podcast.

- **Reddit:** Join a cybersecurity sub, like r/netsec, r/cybersecurity, or r/hacking.
- CISA Alerts: Sign up to receive technical alerts from the Cybersecurity & Infrastructure Security Agency.

## Build competitive cybersecurity skills on Coursera

Whether you want to develop a new skill, get comfortable with an in-demand technology, or advance your abilities, keep growing with a Coursera Plus subscription. You'll get access to over 10,000 flexible courses from over 350 top universities and companies.



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## Frequently asked questions (FAQs)

Is coding required for cybersecurity?
Does cybersecurity require a lot of math?
What human skills are required for cybersecurity?

## Best Cybersecurity Analysis courses for beginners

Top-rated beginner-friendly Cybersecurity Analysis courses with no prerequisites.

1. 1
   Microsoft Cybersecurity Analyst
   MicrosoftBeginner3 - 6 MonthsProfessional Certificate4.7(3,039)
2. 2
   Google Cybersecurity
   GoogleBeginner3 - 6 MonthsProfessional Certificate4.8(68,354)
3. 3
   Introduction to Cybersecurity Fundamentals
   CourseraBeginner1 - 4 WeeksCourse4.6(934)

## Best free Cybersecurity Analysis courses

High-quality free Cybersecurity Analysis courses you can start today.

1. 1
   CompTIA Security+ 701
   InfosecIntermediate3 - 6 MonthsCourse4.6(732)
2. 2
   Global Systemic Risk
   Princeton UniversityBeginner1 - 3 MonthsCourse4.7(457)
3. 3
   Algorithms, Part II
   Princeton UniversityIntermediate3 - 6 MonthsCourse4.9(2,050)

## Best Cybersecurity Analysis professional programs

Earn a certificate in Cybersecurity Analysis from top universities and companies.

1. 1
   Cybersecurity Analyst
   EdurekaIntermediate3 - 6 MonthsSpecialization4.7(6)
2. 2
   IBM Cybersecurity Analyst
   IBMBeginner3 - 6 MonthsProfessional Certificate4.6(28,294)
3. 3
   Cybersecurity in modern organizations
   CourseraIntermediate1 - 3 MonthsSpecialization4.6(1,157)

## Article sources

1. GIAC Certifications. "Top 5 Cloud Certs You Need to Know About, https://www.giac.org/blog/top-five-cloud-certs/." Accessed October 9, 2025.

## Keep reading

- What Does a Cryptanalyst Do? 2026 Career Guide
  6 min read · October 22, 2025
- What Is the CompTIA Security+ (Plus) Certification? 2026 Guide
  8 min read · March 27, 2026
- Information Security Analyst: What They Do and How to Become One
  October 30, 2025
- 8 Popular Cybersecurity Certifications in 2026
  11 min read · August 7, 2026

- Cybersecurity Degrees and Alternatives: Your 2026 Guide
  7 min read · July 16, 2026
- Cybersecurity Frequently Asked Questions (FAQ)
  May 7, 2026

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- Artificial Intelligence (AI)
- Cybersecurity
- Data Analytics
- Digital Marketing
- Human Resources (HR)
- Microsoft Excel

- Project Management
- Python
- SQL

## Professional Certificates

- Google AI Certificate
- Google Cybersecurity Certificate
- Google Data Analytics Certificate
- Google IT Support Certificate
- Google Project Management Certificate
- Google UX Design Certificate
- IBM AI Engineering Certificate
- IBM AI Product Manager Certificate
- IBM Data Science Certificate
- Intuit Academy Bookkeeping Certificate

## Courses & Specializations

- AI Essentials Specialization
- AI For Business Specialization
- AI For Everyone Course
- AI in Healthcare Specialization
- Deep Learning Specialization
- Excel Skills for Business Specialization
- Financial Markets Course
- Machine Learning Specialization
- Prompt Engineering for ChatGPT Course
- Python for Everybody Specialization

## Career Resources

- Career Aptitude Test
- CAPM Certification Requirements
- CompTIA A+ Certification Requirements
- CompTIA Security+ Certification Requirements
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- High-Income Skills to Learn
- How to Learn Artificial Intelligence
- PMP Certification Requirements
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